`microimpute`: Benchmarking Cross-Survey Imputation Methods for U.S. Household Wealth

María Juaristi*

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Abstract

Microdata surveys often lack variables needed for policy analysis, requiring imputation from richer donor surveys, yet no standardized tools exist for comparing imputation methods and selecting the best approach for a given dataset. We introduce `microimpute`, an open-source Python package for benchmarking, tuning, and automatically selecting among multiple imputation methods. We apply it to impute U.S. household wealth from the Survey of Consumer Finances (SCF) onto the Current Population Survey (CPS). Evaluating five methods (Quantile Random Forests, Ordinary Least Squares, Quantile Regression, Hot Deck Matching, and Mixture Density Networks) through a 5-fold cross-validation, we find that QRF reduces average quantile loss by 76% relative to the worst-performing method and by 7% relative to the next-best method (Hot Deck Matching), owing to its ability to model non-linear relationships between demographic predictors and wealth. Its Wasserstein distance to the donor distribution is 24% lower than Matching's. A simulation of the Supplemental Security Income Savings Penalty Elimination Act shows what is at stake: without imputed wealth, the microsimulation overestimates SSI recipients by 167%, and the reform is entirely un-simulable. Cross-dataset benchmarking across six additional domains, however, revealsthat no single method dominates universally. QRF performs best when relationships are non-linear, Hot Deck Matching better preserves marginal distributions by drawing directly from the donor pool, and OLS remains competitive when relationships are approximately linear. Imputation method selection should therefore be empirically driven rather than prescribed, which motivates tools like `microimpute` for systematic comparison.

1 Introduction

Statistical matching, also known as data fusion or full variable imputation, combines information from multiple data sources that share common variables but contain different samples of units (D'Orazio et al., 2006), a situation common across empirical research fields. In its

*Minerva University, PolicyEngine

simplest form, the problem involves a donor dataset containing both shared and unique variables and a receiver dataset containing only the shared variables, where the goal is to impute the missing variables for observations in the receiver file based on their shared characteristics (D'Orazio et al., 2021). This framework generalizes traditional missing data imputation by recognizing that "missingness" can arise not only from nonresponse within a single survey but also from the structural absence of variables across distinct data sources.

Statistical matching is widely applicable in the social sciences, particularly in microsimulation modeling, where policy analysis often requires combining detailed demographic information from one survey with economic variables from another (Bourguignon and Spadaro, 2006). For instance, household surveys may capture income and consumption patterns but lack wealth data, while financial surveys provide wealth information for different samples. Constructing synthetic files that combine these variables allows more detailed policy analysis than either source alone.

The main methodological difficulty is accurately modeling the conditional distribution of the target variable given shared predictors, as different approaches make different assumptions about this relationship. Parametric methods may be too restrictive for economic variables with heavy tails and heteroscedasticity, while nonparametric and machine learning approaches are more flexible but also more complex (Meinshausen, 2006; Andridge and Little, 2010). Section 2 reviews these trade-offs in detail.

This paper presents the `microimpute` package, a Python library implementing five statistical matching methods: Hot Deck Matching, Ordinary Least Squares, Quantile Regression, Mixture Density Networks, and Quantile Random Forests. The package provides a framework for comparing these methods through systematic benchmarking, with automated model selection based on quantile loss performance. We demonstrate the methodology through an application to wealth imputation, transferring net worth data from the Survey of Consumer Finances to the Current Population Survey. This task exemplifies the challenges of statistical matching for heavy-tailed distributions.

This work provides the following contributions:

- An open-source implementation of five imputation methods under a common interface, allowing systematic comparison across methods
- An automated benchmarking pipeline that evaluates distributional accuracy using quantile loss, so that researchers can select the most appropriate method for their data
- Empirical evidence on the relative performance of these methods for wealth imputation, suggesting that Quantile Random Forests are well-suited to heavy-tailed distributions, though method choice remains context-dependent
- A policy reform simulation of the SSI Savings Penalty Elimination Act, showing that imputation method choice produces meaningfully different estimates of reform impact, and that without imputed wealth data, such reforms cannot be simulated at all

The remainder of this paper is organized as follows. Section 2 reviews the statistical matching problem, including its formal definition, the conditional independence assumption that underlies all matching methods, and the five imputation approaches implemented in `microimpute`. Section 3 describes our data sources. Section 4 presents the package architecture and benchmarking methodology. Section 5 reports empirical results from the wealth imputation application. Section 6 discusses implications and limitations, and Section 7 concludes.

2 Background

This section defines the statistical matching problem and describes the imputation methods in `microimpute`. We begin with the formal problem definition and the assumption underlying all statistical matching procedures, then discuss the relationship to missing data mechanisms, and finally describe each of the five imputation methods.

2.1 The Statistical Matching Problem

Statistical matching arises when a researcher seeks to combine information from two (or more) data sources that have been collected independently on different samples (D'Orazio et al., 2006). The canonical setup involves:

- A **donor file** $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n_D}$ containing n_D observations with common variables X and target variable(s) Y
- A **receiver file** $\mathcal{R} = \{x_j\}_{j=1}^{n_R}$ containing n_R observations with only the common variables X

The goal is to impute values $\hat{y}_j$ for each observation j in the receiver file, creating a synthetic file that combines the variables from both sources. When the receiver file also contains variables Z not present in the donor file, the resulting synthetic file contains all three sets of variables (X, Y, Z), though Y and Z are never jointly observed.

Formally, statistical matching seeks to estimate the conditional distribution $P(Y|X)$ from the donor data and use it to generate imputations for the receiver observations. Let $\hat{P}(Y|X = x)$ denote an estimate of this conditional distribution. For a receiver observation with covariates x_j, we can either:

1. Predict a specific quantile, such as the conditional median: $\hat{y}_j = \hat{Q}_{0.5}(Y|X = x_j)$
2. Draw a random value from the estimated distribution: $\hat{y}_j \sim \hat{P}(Y|X = x_j)$

The first approach (deterministic imputation) may be appropriate when only point predictions are needed, though it systematically underestimates variance (Rubin, 1987). The second approach (stochastic imputation) preserves the variability of Y and is preferred when distributional properties must be maintained. This is often the case in microsimulation modeling, where full distributional policy analysis is required.

2.2 The Conditional Independence Assumption

All statistical matching procedures rely, either explicitly or implicitly, on the Conditional Independence Assumption (CIA):

$$Y \perp Z \mid X \tag{1}$$

This assumption states that the target variable Y (from the donor) and any variables Z unique to the receiver are independent, conditional on the common variables X (D'Orazio et al., 2006). In other words, once we account for the shared covariates, knowing Z provides no additional information about Y.

The CIA is fundamentally untestable because Y and Z are never jointly observed in either data source. Its plausibility depends on the richness of the common variables X. When X captures the key determinants of both Y and Z, the assumption is more likely to hold. In practice, researchers must rely on substantive knowledge to assess whether the available common variables are sufficient to render Y and Z conditionally independent.

Violations of the CIA lead to biased estimates of the joint distribution of (Y, Z) in the synthetic file. However, if the research question concerns only the marginal distribution of Y in the receiver population, the CIA is not strictly required. What matters is that the conditional distribution $P(Y|X)$ is correctly specified and that X has sufficient overlap between donor and receiver files (D'Orazio et al., 2021).

2.3 Missing Data Mechanisms

Statistical matching may also warrant discussion within the framework of missing data theory, where Y is "missing" for all observations in the receiver file (Little and Rubin, 2002). The missing data literature distinguishes three mechanisms:

- **Missing Completely at Random (MCAR)**: The probability of missingness is unrelated to both observed and unobserved variables. In the statistical matching context, this would require that selection into the donor versus receiver sample is independent of all variables.
- **Missing at Random (MAR)**: The probability of missingness depends only on observed variables. For statistical matching, this means that conditional on X, the probability of being in the donor sample (and thus having Y observed) does not depend on the value of Y itself.
- **Missing Not at Random (MNAR)**: The probability of missingness depends on the unobserved values. This would occur if, for example, high-wealth individuals were systematically less likely to appear in the wealth survey.

Under MAR, valid inference about $P(Y|X)$ can be obtained from the donor data alone, which justifies the statistical matching approach. Most imputation methods, including those implemented in `microimpute`, assume MAR. When MNAR is suspected, sensitivity analyses or methods that explicitly model the selection mechanism may be required (Little and Rubin, 2002).

In statistical matching applications, an important consideration arises when the donor and receiver files come from surveys with different sampling designs. For instance, wealth surveys often oversample high-net-worth households to improve precision for this rare population, while general household surveys use designs representative of the broader population. This differential sampling can introduce a form of selection that resembles MNAR if not properly addressed. Luckily, survey weights provide a mechanism to mitigate this issue. By incorporating weights that reflect each observation's representativeness of the target population, imputation methods can adjust for known differences in sampling probabilities between the donor and receiver surveys (DuGoff et al., 2014). The `microimpute` package supports the use of survey weights throughout its imputation pipeline, allowing methods to account for complex survey designs. While survey weights cannot fully address MNAR when missingness depends on unobserved values of Y itself, they can substantially reduce bias arising from differential sampling designs across data sources.

2.4 Imputation Methods

We now describe the five imputation methods implemented in `microimpute`. Each method provides a different approach to estimating the conditional distribution $P(Y|X)$ and generating imputations.

2.4.1 Hot Deck Matching

Hot deck imputation replaces missing values in a receiver record with observed values from "similar" donor records (Andridge and Little, 2010). The `microimpute` implementation uses the unconstrained distance hot deck approach from the StatMatch R package (D'Orazio et al., 2021). To identify the best match, the method computes distances between each receiver observation and all donor observations based on the common variables X, selects donor records within a specified distance threshold, and randomly samples from the eligible donors, with optional weighting by survey weights. For continuous and mixed covariates, Mahalanobis or Gower distance metrics can capture similarity across different variable types. The donated value is therefore always an actual observed value from the donor file.

Hot deck methods are nonparametric and avoid distributional assumptions, making them robust when the true conditional distribution is unknown or complex (D'Orazio et al., 2006). However, they face limitations:

- **Donor scarcity**: For receiver observations in sparse regions of the covariate space suitable donors may be unavailable, which can lead to the reuse of the same donors, which can affect the accuracy in the distribution of imputed values

- **Discrete outputs**: Imputed values are restricted to the set of observed donor values, which may not adequately represent the continuous nature of Y

- **Tail behavior**: Extreme values in the receiver population may have no appropriate donors or poor matches, leading to biased imputations at the tails

2.4.2 Ordinary Least Squares (OLS)

OLS imputation models the conditional mean of Y given X through linear regression:

$$E[Y|X] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p \tag{2}$$

The coefficients $\boldsymbol{\beta}$ are estimated from the donor data by minimizing squared residuals. For stochastic imputation, the `microimpute` implementation, which builds on the `statsmodels` library, assumes normally distributed errors:

$$Y|X \sim \mathcal{N}(\boldsymbol{X}'\hat{\boldsymbol{\beta}}, \hat{\sigma}^2) \tag{3}$$

where $\hat{\sigma}^2$ is the estimated residual variance. Quantiles of $Y|X$ can then be computed as:

$$\hat{Q}_\tau(Y|X) = \boldsymbol{X}'\hat{\boldsymbol{\beta}} + \hat{\sigma} \cdot \Phi^{-1}(\tau) \tag{4}$$

where $\Phi^{-1}(\tau)$ is the τ-th quantile of the standard normal distribution.

OLS is computationally efficient and well-understood, but its assumptions are often violated by economic variables:

- **Linearity**: Relationships between Y and X may be nonlinear
- **Homoscedasticity**: Variance often increases with the level of Y, particularly for wealth
- **Normality**: Heavy-tailed distributions violate the normal error assumption, leading to inaccurate quantile estimates (Von Hippel, 2007)

2.4.3 Quantile Regression

Quantile regression directly models conditional quantiles rather than the conditional mean (Koenker and Bassett, 1978). For quantile $\tau \in (0, 1)$, the model is:

$$Q_\tau(Y|X) = \beta_0(\tau) + \beta_1(\tau)x_1 + \cdots + \beta_p(\tau)x_p \tag{5}$$

The coefficients $\boldsymbol{\beta}(\tau)$ are estimated by minimizing the asymmetric quantile loss function:

$$\mathcal{L}_\tau(\boldsymbol{\beta}) = \sum_{i=1}^{n} \rho_\tau(y_i - \boldsymbol{x}_i'\boldsymbol{\beta}) \tag{6}$$

where the check function $\rho_\tau(u) = u(\tau - \mathbf{1}_{u<0})$ penalizes positive and negative residuals asymmetrically.

The `microimpute` implementation of Quantile regression also uses the `statsmodels` library. By fitting models at multiple quantiles, it can approximate the entire conditional distribution of $Y|X$. For imputation, a random quantile $\tau^* \sim \text{Uniform}(0, 1)$ can be drawn, and the imputed value computed by interpolating between estimated quantile functions (Wei et al., 2014).

Quantile regression offers advantages over OLS:

- **Robustness**: Less sensitive to outliers than least squares
- **Heteroscedasticity**: Naturally accommodates varying spread across the distribution
- **No normality assumption**: Does not require specifying an error distribution

However, quantile regression still assumes linear relationships at each quantile, limiting its flexibility for complex data structures (Meinshausen, 2006).

2.4.4 Mixture Density Networks (MDN)

Mixture Density Networks combine neural networks with mixture models to estimate flexible conditional densities (Bishop, 1994). The approach models $P(Y|X)$ as a mixture of Gaussian components:

$$p(y|x) = \sum_{k=1}^{K} \pi_k(x) \cdot \mathcal{N}(y \mid \mu_k(x), \sigma_k^2(x)) \tag{7}$$

where K is the number of mixture components, and the mixing coefficients $\pi_k(x)$, means $\mu_k(x)$, and variances $\sigma_k^2(x)$ are all functions of the input x, parameterized by a neural network.

The neural network outputs $3K$ values for each input: K unnormalized log-probabilities (transformed via softmax to obtain π_k), K means, and K log-variances (exponentiated to ensure positivity). The network is trained by maximizing the log-likelihood:

$$\mathcal{L} = \sum_{i=1}^{n} \log p(y_i|x_i) \tag{8}$$

For imputation, one can either sample from the mixture distribution or compute quantiles numerically by inverting the cumulative distribution function.

The `microimpute` implementation uses PyTorch Tabular (Joseph, 2021), which provides optimized architectures for tabular data. MDNs offer several advantages:

- **Multimodal distributions**: The mixture formulation can capture multiple modes in $P(Y|X)$
- **Flexible nonlinearity**: Neural networks can approximate complex relationships between X and distributional parameters
- **Full density estimation**: Provides a complete probabilistic model, not just quantiles

Nonetheless, its limitations include:

- **Data requirements**: Neural networks typically require larger sample sizes for reliable estimation
- **Hyperparameter sensitivity**: Performance depends on architecture choices, learning rate, and number of components
- **Computational cost**: Training is more expensive than parametric methods

2.4.5 Quantile Random Forests (QRF)

Quantile Random Forests extend Random Forests to estimate conditional quantiles (Meinshausen, 2006). A standard Random Forest builds an ensemble of decision trees, each trained on a bootstrap sample, with predictions averaged across trees. QRF modifies this by retaining all training observations in each terminal node rather than just their mean.

For a new observation with covariates x, QRF estimates the conditional distribution function as:

$$\hat{F}(y|X = x) = \sum_{i=1}^{n} w_i(x) \cdot \mathbf{1}_{Y_i \leq y} \tag{9}$$

where the weight $w_i(x)$ reflects how often training observation i falls in the same terminal node as x across trees in the forest. Specifically:

$$w_i(x) = \frac{1}{B} \sum_{b=1}^{B} \frac{\mathbf{1}_{x_i \in L_b(x)}}{|L_b(x)|} \tag{10}$$

where B is the number of trees, $L_b(x)$ is the terminal node containing x in tree b, and $|L_b(x)|$ is the number of training observations in that node.

The τ-th conditional quantile is then:

$$\hat{Q}_\tau(Y|X = x) = \inf\{y : \hat{F}(y|X = x) \geq \tau\} \tag{11}$$

The `microimpute` implementation uses the `quantile-forest` package, which offers several advantages for statistical matching:

1. **Distribution preservation**: By modeling the entire conditional distribution, QRF captures skewness, heavy tails, and other distributional features without parametric assumptions (Meinshausen, 2006).

2. **Nonlinear relationships**: The tree-based structure automatically captures complex nonlinearities and interactions without requiring explicit specification (Breiman, 2001).

3. **Single model for all quantiles**: Unlike standard quantile regression, which requires fitting separate models for each quantile, QRF estimates all quantiles from a single trained forest.

4. **Robustness**: The ensemble approach and quantile focus make QRF less sensitive to outliers than mean-based methods (Tang and Ishwaran, 2017).

Nevertheless, QRF also has limitations:

- **Extrapolation**: Trees cannot extrapolate beyond the range of training data, potentially underestimating extreme quantiles when donor and receiver distributions differ

- **Discrete approximation**: The estimated distribution is discrete, with support limited to observed Y values in the training data

2.5 Current Practice in Microsimulation

Statistical matching and data fusion are fundamental operations in microsimulation modeling, yet the methods employed in practice have remained relatively unchanged for decades. A review of major tax-benefit microsimulation models reveals a strong reliance on traditional imputation approaches, primarily Hot Deck Matching and OLS-based regression.

Hot Deck Matching remains the dominant approach in European microsimulation. EUROMOD, the EU-wide tax-benefit model, employs a multi-stage imputation procedure combining predictive mean matching with distance-based hot deck methods to integrate consumption data from Household Budget Surveys into its EU-SILC input data (Sutherland and Figari, 2013). Hot deck methods are simple to implement and guarantee that imputed values are actual observed values from the donor file. However, as discussed above, these methods struggle with tail behavior and may not adequately capture the full conditional distribution of the target variable.

In U.S. tax policy microsimulation, regression-based approaches are more common. The Tax Policy Center employs a two-stage probit and OLS procedure for wealth imputation, first predicting the probability of holding each asset type, then predicting amounts conditional on positive holdings (Nunns, 2012). Similarly, the Institute on Taxation and Economic Policy (ITEP) model relies on statistical matching between tax return data and the American Community Survey, supplemented with regression-based imputations from the Survey of Consumer Finances (Institute on Taxation and Economic Policy, 2023). These regression approaches assume linear relationships and normally distributed errors, assumptions frequently violated by economic variables with heavy tails and heteroscedasticity.

More recent microsimulation efforts have begun incorporating administrative data through record linkage rather than statistical matching (Abowd, 2018), but this approach requires access to restricted data and raises privacy concerns. For researchers working with publicly available survey data, statistical matching remains the primary option, and there is room for methods that capture distributional features more effectively.

These limitations are especially pronounced for heavy-tailed variables where the relationship with predictors varies across the distribution. Machine learning methods such as Quantile Random Forests can capture nonlinear relationships, model the entire conditional distribution, and handle heavy-tailed data without restrictive parametric assumptions, yet they have seen limited adoption in microsimulation practice. The `microimpute` package provides a common framework for comparing traditional and machine learning approaches, so that researchers can empirically evaluate which method works best for their data.

2.6 Supplemental Security Income as a Validation Case

The limitations of traditional imputation approaches have concrete consequences for downstream policy analysis. The Supplemental Security Income (SSI) program illustrates this well. SSI is a federal means-tested program for aged, blind, and disabled individuals with strict resource limits ($2,000 for individuals, $3,000 for couples) that have remained unchanged since 1989 (Social Security Administration, 2023). The resource test requires household wealth and asset data that the CPS does not collect, making SSI eligibility determination dependent on imputed or heuristic wealth values.

In 2024, SSI served approximately 7.4 million recipients with $59.6 billion in federal payments (Social Security Administration, 2024). Because different wealth imputation methods produce different wealth distributions, they directly affect which simulated households pass the resource test and therefore the resulting estimates of SSI recipient counts and expenditure. This makes SSI a particularly informative case for comparing imputation approaches, as described in Section 4.4.

The frozen resource limits also make SSI a compelling case for policy reform analysis. In 2024 dollars, the original 1989 thresholds of $2,000/$3,000 are worth approximately $4,800/$7,200, meaning the resource test has become substantially more restrictive over time without any legislative change. The SSI Savings Penalty Elimination Act (S. 1234 / H.R. 2540), a bipartisan bill introduced in April 2025, proposes raising these limits to $10,000 for individuals and $20,000 for couples, with future CPI indexing (119th United States Congress, 2025). Simulating the impact of this reform, and determining which additional households would gain eligibility under the higher thresholds requires household-level wealth data, making it a natural application for cross-survey imputation. As we show in Section 4.4, the choice of imputation method produces meaningfully different estimates of this reform's impact, meaning that imputation is not merely a data preprocessing step but a modeling choice with real consequences for policy analysis.

3 Data

The selected application to demonstrate the `microimpute` methodology involves transferring net worth from the Survey of Consumer Finances (SCF) onto the Current Population Survey (CPS).

3.1 Survey of Consumer Finances

The Survey of Consumer Finances, sponsored by the Federal Reserve Board, is a triennial survey providing detailed information on U.S. households' assets, liabilities, income, and demographic characteristics. Its dual-frame sample design includes a standard national area-probability sample and a list sample that deliberately oversamples wealthy households to better capture the skewed wealth distribution (Barceló, 2006). The SCF is a standard reference for wealth imputation research because of its detailed financial data and the known complexities arising from its design and the nature of wealth. Item nonresponse in public-use SCF datasets is addressed by the Federal Reserve through a multiple imputation approach that generates five complete datasets with different imputed values, using sequential regression-based procedures that incorporate range constraints, logical data structures, and empirical residuals to preserve the complex multivariate relationships inherent in wealth data (Kennickell, 1998).

Specifically, we use the 2022 summarized SCF as our donor dataset.

3.2 Current Population Survey

The Current Population Survey, conducted jointly by the U.S. Census Bureau and the Bureau of Labor Statistics, is a monthly survey of approximately 60,000 U.S. households and is the primary source of labor force statistics for the United States. The CPS uses a multistage probability-based sample designed to represent the civilian non-institutional population; on average, each sampled household represents approximately 2,500 households in the population. The Annual Social and Economic Supplement (ASEC) extends the core survey with detailed annual income data, including earnings, unemployment compensation, Social Security, pension income, interest, dividends, and other income sources. However, despite its broad coverage of income and employment, the CPS does not collect information on household wealth, assets, or liabilities, which are needed for wealth-based policy analysis. This omission motivates the need for statistical matching to transfer wealth information from the SCF onto the CPS.

Specifically, we use the Enhanced CPS 2024 produced by PolicyEngine (PolicyEngine, 2024) as our receiver dataset. The Enhanced CPS extends the base CPS ASEC microdata through survey reweighting and calibration to administrative totals, improving its representativeness for tax-benefit microsimulation. The SCF 2022 financial variables are uprated to 2024 dollars using CPI adjustment factors to ensure consistency with the receiver dataset's reference year.

3.3 Comparative analysis and characteristics for imputation

The SCF and CPS differ fundamentally in sampling design: the SCF employs a dual-frame approach combining a standard area-probability sample with a list sample that deliberately oversamples wealthy households, while the CPS uses a multistage area-probability household design representative of the civilian non-institutional population. This difference means that the donor and receiver surveys weight different parts of the wealth distribution differently, making proper survey weight integration during model training essential for unbiased imputation. The two surveys share a core set of demographic and income variables (age, sex, race, number of children, employment income, interest and dividend income, and pension income) but differ in detailed coverage: the SCF collects granular asset and liability data absent from the CPS, while the CPS captures labor force dynamics and program participation variables not available in the SCF. This partial overlap constrains the conditioning set available for imputation and reinforces the importance of the conditional independence assumption discussed in Section 2.2. The contrasting survey designs, the heavy-tailed nature of wealth, and the limited variable overlap make the SCF-to-CPS imputation an informative test case for comparing methods. It is also directly relevant to U.S. policy microsimulation, where analyzing the distributional effects of tax and benefit reforms requires microdata that combine income, demographics, and wealth.

4 Methodology

4.1 `microimpute` package implementation

`microimpute`[1] is an open-source Python framework for statistical matching that provides a common interface for imputing variables across survey datasets using multiple methods. While several imputation approaches exist individually, researchers have lacked a standardized tool for comparing their distributional performance on a given dataset and automatically selecting the best approach. In this study, `microimpute` is both a methodological contribution and the analytical engine behind the wealth imputation experiments described below.

4.1.1 Core capabilities

The package currently supports five imputation methods: Hot Deck Matching, Ordinary Least Squares Linear Regression (OLS), Quantile Regression, Quantile Random Forests (QRF), and Mixture Density Networks (MDN). The package is modular, so additional imputation methods can be added without modifying the existing codebase.

4.1.2 Key features

Several features are relevant to the cross-survey imputation experiments presented in this paper:

1. **Survey weight integration**: Incorporates survey weights through sampling so that models are trained on data representative of the true population distribution.

2. **Method comparison and benchmarking**: Compares different approaches and automatically identifies the method with the lowest quantile loss.

3. **Hyperparameter tuning**: Supports user-specified hyperparameters as well as automated tuning.

4. **Quantile-based evaluation**: Assesses imputation quality across different parts of the distribution using quantile loss.

5. **Autoimputation**: Tunes hyperparameters, compares methods, and selects the best-performing one to impute onto the receiver dataset, all in a single function call.

4.2 Evaluation framework

Comparing imputation methods requires metrics that assess distributional accuracy, not just point prediction. We employ two complementary quantitative metrics, along with visual distributional assessments.

[1] Documentation, implementation details, and usage examples are available at https://policyengine.github.io/microimpute/.

The primary evaluation metric is quantile loss (also known as pinball loss). For a target quantile $\tau \in (0, 1)$, quantile loss assigns an asymmetric penalty that depends on the sign of the forecast error:

$$\mathcal{L}_\tau = \rho_\tau(y - \hat{Q}_\tau(y|x)) = (y - \hat{Q}_\tau(y|x)) \cdot (\tau - \mathbf{1}_{y<\hat{Q}_\tau(y|x)}) \tag{12}$$

Under-prediction of an upper-tail quantile is penalized at rate τ, whereas over-prediction is penalized at rate $1 - \tau$. This asymmetric structure captures directional bias, which matters when imputing highly skewed variables like wealth, where large under-estimates in the right tail must be discouraged more strongly than equal-sized over-estimates (Koenker and Bassett, 1978). Compared with mean-squared or mean-absolute error, which penalize errors symmetrically, quantile loss directly targets the conditional distribution and remains robust to outliers (Ghenis, 2018). Averaging over multiple quantiles $\tau \in \{0.1, 0.2, \ldots, 0.9\}$ summarizes how well the method estimates the conditional distribution across its entire range. Lower average quantile loss indicates better distributional calibration. Because quantile loss is not scale-invariant, the absolute values reported in Section 5 reflect the scale transformation applied; we supplement raw values with relative improvement percentages to aid interpretability.

As a complementary metric, we use the Wasserstein distance (also known as the earth mover's distance) to measure overall distributional similarity between imputed and reference distributions. Formally, for one-dimensional distributions P and Q:

$$W_1(P, Q) = \int_0^1 |F_P^{-1}(u) - F_Q^{-1}(u)| \, du \tag{13}$$

where F_P^{-1} and F_Q^{-1} are the quantile functions. The Wasserstein distance captures the total "work" required to transform one distribution into another, making it sensitive to the magnitude of distributional differences. While quantile loss evaluates conditional predictions at specific quantiles, the Wasserstein distance provides a single summary measure of marginal distributional accuracy, which is particularly informative for heavy-tailed variables like wealth.

In addition to these quantitative metrics, we employ two visual assessment approaches. First, we compare the imputed wealth distributions against the weighted SCF donor distribution, which, given the SCF's household weights that account for survey representativeness, provides a reference for the true U.S. wealth distribution. Second, we evaluate the average imputed wealth by disposable income decile, assessing whether each method produces a plausible monotonically increasing wealth-income relationship consistent with economic expectations.

4.3 Experimental setup

The `autoimpute` function orchestrates the full imputation pipeline in a single call: it tunes hyperparameters for applicable methods using the Optuna framework (Akiba et al., 2019), evaluates all methods through k-fold cross-validation on the donor data, and automatically selects the best-performing one to impute onto the receiver dataset. Using this pipeline, we evaluate all five imputation methods (QRF, OLS, Quantile Regression, Hot Deck Matching, and MDN) for net worth imputation from the SCF to the CPS.

Prior to model training, we apply an inverse hyperbolic sine (asinh) transformation to the net worth target variable. The asinh function, $\text{asinh}(x) = \ln(x + \sqrt{x^2 + 1})$, approximates the natural logarithm for large positive values while remaining well-defined for zero and negative values. This is particularly important for wealth data, where a meaningful fraction of households report negative net worth (i.e., debts exceeding assets), making a standard logarithmic transformation infeasible. The transformation compresses the extreme right tail of the wealth distribution, stabilizing variance and improving model performance across all methods. After imputation, predictions are back-transformed via the inverse function $\sinh(\cdot)$ to recover values on the original dollar scale.

This experimental design implicitly assumes that wealth is missing at random (MAR) conditional on the shared predictors X, i.e., that the probability of observing wealth in the donor survey does not depend on wealth itself once we condition on the demographic and financial variables available in both datasets, as formalized in Section 2.2. This assumption is standard in statistical matching but may be violated in practice: wealthy households may be less likely to participate in surveys or may underreport assets, creating MNAR-like patterns that the shared predictors cannot fully capture. We return to this limitation and discuss partial sensitivity checks in Section 6.4.

To create a ground truth for evaluation, we employ cross-validation, splitting the SCF into 5 folds. For each method, `autoimpute` performs the following steps:

1. Mask the wealth values of one holdout fold at a time
2. Tune the hyperparameters of methods with flexible configurations to the SCF dataset, namely Matching[2], QRF[3], and MDN[4]
    (a) Hyperparameter tuning is conducted through the `optuna` package, which parallelizes hyperparameter searches to identify the combination minimizing the average quantile loss across all quantiles for the SCF dataset
3. Impute wealth (`networth`) onto the SCF holdout set using each method, with the remaining four folds as training data
    (a) SCF survey data weights (captured in the variable `household_weight`) are passed into `autoimpute` for sampling
    (b) Demographic and financial variables available in both surveys serve as predictors, after preprocessing for consistent variable names and encoding: `is_female`, `age`, `race`, `own_children_in_household`, `employment_income`, `interest_dividend_income`, `pension_income`
    (c) Each method performs imputation at 20 equally spaced quantiles

[2]Hyperparameters: `dist_fun`, `k`.

[3]Hyperparameters: `n_estimators`, `min_samples_split`, `min_samples_leaf`, `max_features`, `bootstrap`.

[4]Hyperparameters: `num_gaussian`, `learning_rate`.

4. Compare the imputed values to the SCF holdout values, measuring quantile loss at each of the imputation quantiles
5. Average quantile loss across all folds and quantiles
6. Select the method with the lowest average quantile loss as the best performer
7. Retrain the selected method on the full SCF dataset and use it to perform the final imputation of net worth onto the CPS

This end-to-end pipeline allows for direct comparison between methods on a common evaluation framework and provides a robust assessment of relative performance. The final imputation step uses the entire SCF as training data, maximizing the information available to the selected model.

The imputation from the SCF onto the CPS is then replicated for the remaining four methods with an identical setup: training on the full SCF, the same predictor variables, and imputations at the median quantile to avoid introducing bias toward a specific side of the distribution.

4.4 Downstream policy analysis

While distributional metrics such as quantile loss and Wasserstein distance assess how well an imputation method recovers the statistical properties of the target variable, they do not directly capture whether differences between methods matter for downstream policy analysis. To address this, we introduce a policy analysis layer that evaluates imputation quality through its effect on a concrete policy simulation, under both current law and a proposed reform.

We use the Supplemental Security Income (SSI) program as the policy case. SSI is a federal means-tested program that provides cash assistance to aged, blind, and disabled individuals with limited income and resources (Social Security Administration, 2023). SSI eligibility includes a resource test with limits of \$2,000 for individuals and \$3,000 for couples, which depends on household wealth data that the CPS does not collect.

In current practice, PolicyEngine's microsimulation model (PolicyEngine, 2024) handles the missing wealth data by defaulting `ssi_countable_resources` to zero for all CPS households, meaning every categorically eligible individual passes the resource test. This produces a substantial overestimate of SSI participation. For each imputation method, we feed the model's imputed net worth values into PolicyEngine as `ssi_countable_resources`, replacing the default zero-resource assumption with household-level wealth data. We then simulate SSI eligibility and benefits under each imputation scenario and compare the resulting recipient counts and total expenditure against SSA administrative totals (7.4 million recipients and \$59.6 billion in federal payments for 2024).

Beyond this current-law calibration, we simulate a proposed policy reform: the SSI Savings Penalty Elimination Act (S. 1234 / H.R. 2540), which would raise SSI resource limits from \$2,000/\$3,000 to \$10,000/\$20,000 (119th United States Congress, 2025). For each imputation method, we simulate SSI outcomes under both current law and the reformed thresholds. The difference between the two represents the estimated reform impact.

4.5 Additional methodological analyses

Two supplementary analyses complement the main experimental framework. Appendix B presents a predictor importance analysis using leave-one-out and progressive inclusion experiments on the QRF model, measuring the relative contribution of each predictor to wealth imputation quality and informing the plausibility of the conditional independence assumption. To test whether the relative performance of methods generalizes beyond this particular application, Appendix A reports a cross-dataset benchmarking exercise that evaluates `microimpute`'s five methods across six publicly available datasets from different domains.

5 Results

5.1 Imputation results

The cross-validation results show that QRF performs well across the wealth distribution. With the asinh transformation applied to the net worth target variable, QRF achieved the lowest average quantile loss across all quantiles, outperforming OLS, Hot Deck Matching, Quantile Regression, and MDN. QRF's overall consistency across the entire distribution made it the best-performing method for wealth imputation in this setting.

Automated hyperparameter tuning via the Optuna framework contributed to these results. The optimal QRF configuration identified through cross-validation included 202 trees, a minimum of 1 sample per leaf, a split threshold of 5 samples, approximately 95% of features considered at each split, and bootstrap sampling enabled. These parameters balance model complexity with generalization, preventing overfitting while still capturing non-linear relationships between demographic predictors and wealth.

5.1.1 Quantile loss

Comparing all five models to each other, we can evaluate performance through quantile loss, as well as comparing the overlap of the imputed wealth distribution with the original SCF data.

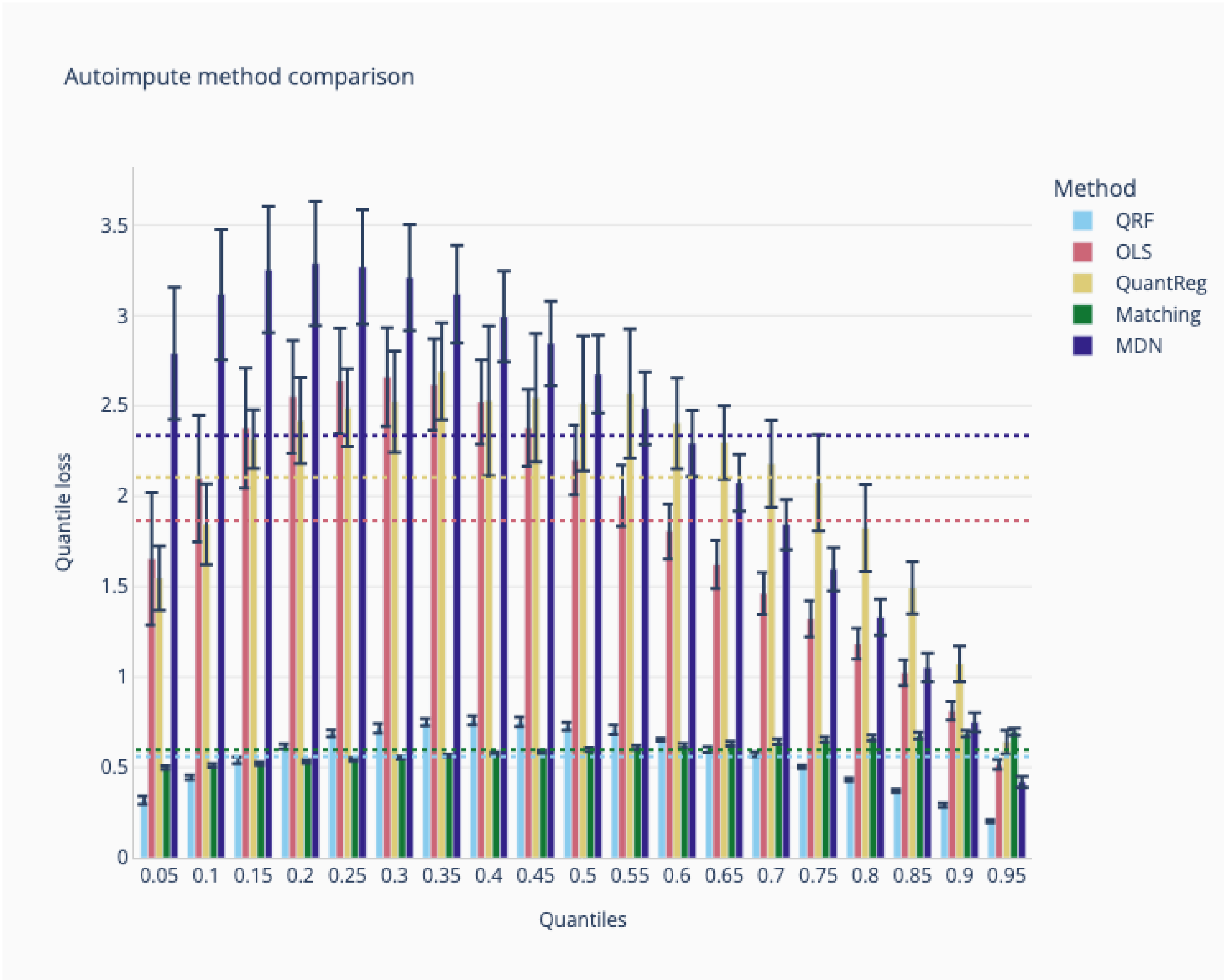

Figure 1: Cross-validation method performance at 20 equally spaced quantiles for all five imputation methods. Test quantile loss at each quantile is averaged across 5 cross-validation folds, and the dashed lines represent average quantile loss across all quantiles.

We observe how QRF proves to be the best-performing model out of the five. It significantly outperforms Matching and OLS on average, while only being outperformed by QuantReg past the 80th quantile. The dashed lines in Figure 1 indicate each method's average quantile loss across all quantiles, clearly separating QRF and Matching from the remaining methods. Matching's performance presents the perfect opportunity to see the quantile loss's directional bias at work. Because Matching does not incorporate quantile information into its imputation process in any way, it will match donor and receiver units, and thus impute values identically regardless of the quantile at work. The laddering behavior observed above results from the fact that Matching's wealth imputations consistently underpredict relative to the true wealth values, and this property is increasingly favored for quantiles below the median, and increasingly penalized as quantiles approach the right end of the distribution.

Table 1 summarizes average quantile loss across all quantiles and cross-validation folds, with relative improvement computed against the worst-performing method (MDN) as a reference.

Table 1: Average quantile loss (asinh scale) across all quantiles and 5-fold cross-validation on the SCF test set. Relative improvement is computed as $(L_{\text{MDN}} - L_{\text{method}})/L_{\text{MDN}} \times 100\%$.

Method	Avg Quantile Loss	Std	Improvement (%)
QRF	0.560	0.016	76.2
Matching	0.601	0.018	74.4
OLS	1.866	0.196	20.6
QuantReg	2.104	0.241	10.4
MDN	2.349	0.227	—

QRF and Matching achieve substantially lower quantile loss than the remaining methods, with QRF reducing loss by 76.2% and Matching by 74.4% relative to MDN. The gap between these two leading methods is modest (0.560 vs. 0.601), while OLS, QuantReg, and MDN cluster at substantially higher loss values.

5.1.2 Imputed wealth distribution comparisons

Comparing the imputed wealth distributions against the weighted donor distribution gives a fuller picture of imputation performance beyond the quantile loss averages measured on the SCF. The distribution of wealth values imputed by QRF closely matches the original SCF distribution, suggesting that QRF performs well not only in quantile loss but also in reproducing the overall shape of the U.S. wealth distribution. Because Matching directly takes values from the donor distribution and imputes them onto the receiver, it is unsurprising that its distribution closely resembles the donor distribution. However, Matching may fail to capture the non-linear relationship between predictors and wealth, producing a marginal distribution that looks correct while assigning inaccurate values to individual households. OLS, QuantReg, and MDN fail to capture variability across the distribution and concentrate imputed values at the lower or higher ends.

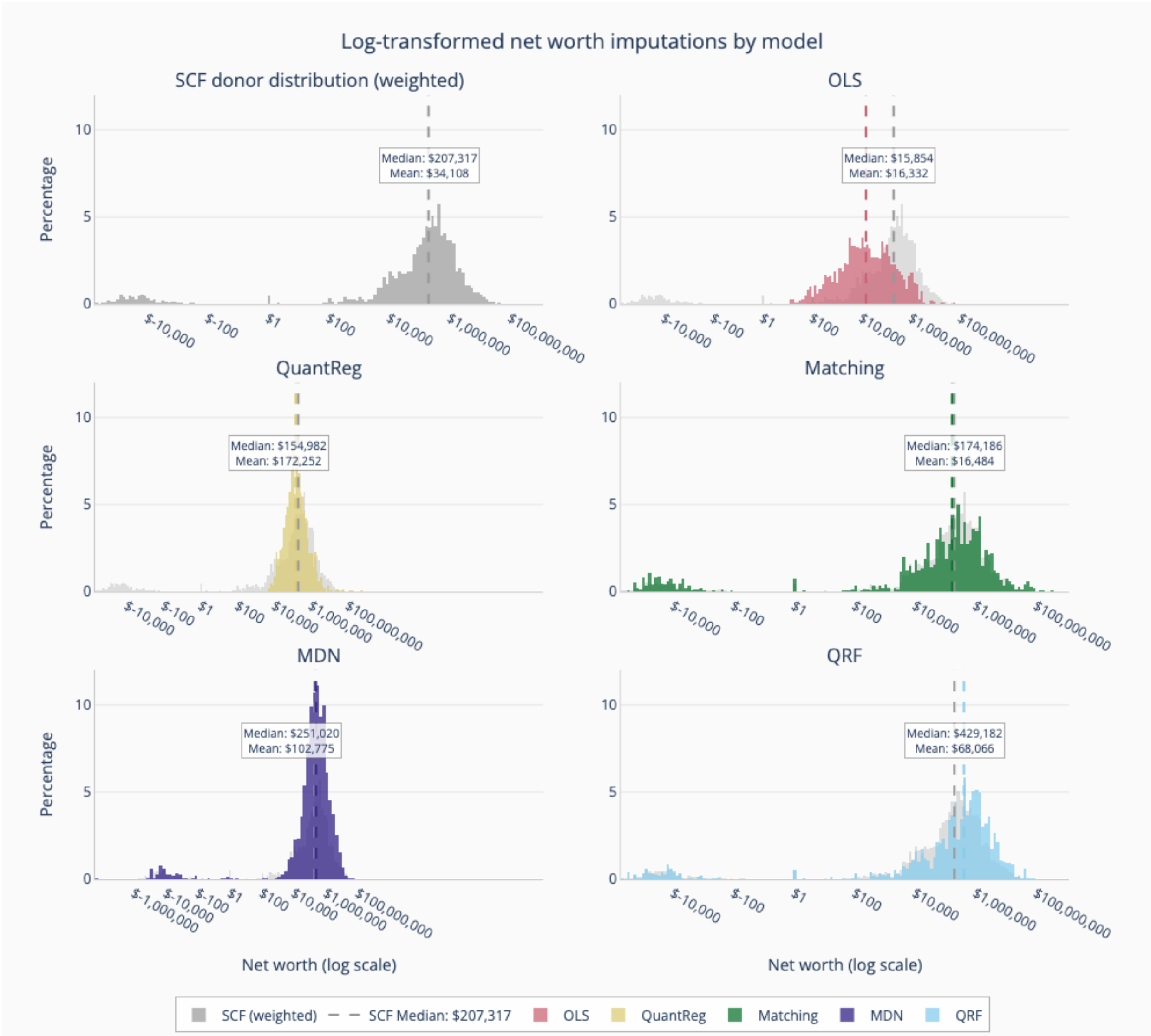

Figure 2: Log-transformed net worth distributions for all five imputation methods, compared to the weighted SCF donor distribution. The y-axis shows the percentage of observations falling in each bin, which normalizes for the different sample sizes between the SCF and CPS and allows direct shape comparison. Dashed lines represent median values.

5.1.3 Wasserstein distance

Table 2 reports the Wasserstein distance between each model's imputed wealth distribution and the weighted SCF donor distribution, providing a single scalar summary of marginal distributional accuracy.

Table 2: Wasserstein distance (earth mover's distance) between each model's imputed net worth distribution on the CPS and the weighted SCF donor distribution. Lower values indicate closer distributional agreement.

Model	Wasserstein Distance
QRF	910,299
Matching	1,192,141
OLS	67,994,870
QuantReg	1.22×10^{14}
MDN	8.23×10^{14}

QRF achieves the lowest Wasserstein distance ($910,299), confirming that its imputed distribution most closely reproduces the donor wealth distribution. To contextualize this magnitude, the 2022 SCF reports a median U.S. household net worth of approximately $192,700 (Board of Governors of the Federal Reserve System, 2023); QRF's Wasserstein distance thus represents roughly 4.7 median household net worths of cumulative distributional displacement, which is small relative to the full range of the wealth distribution but non-trivial in absolute terms. Matching follows at $1.19 million, consistent with its mechanism of drawing values directly from the donor pool. OLS produces a substantially larger distance ($68.0 million), reflecting its compressed predictions that fail to span the full range of the wealth distribution. QuantReg and MDN exhibit Wasserstein distances several orders of magnitude larger, indicating severe distributional mismatch driven by concentrated imputed values in narrow regions.

5.1.4 Distribution of wealth by disposable income decile

Beyond the statistical metrics reported above, we check whether imputed wealth exhibits economically expected relationships with observable variables. Household wealth should increase monotonically with disposable income, given the well-documented positive correlation between income and wealth accumulation. This check matters because a method that achieves competitive quantile loss but produces wealth-income patterns that violate basic economic relationships would be unreliable for downstream policy analysis, where eligibility and benefit calculations depend on the joint distribution of income and wealth.

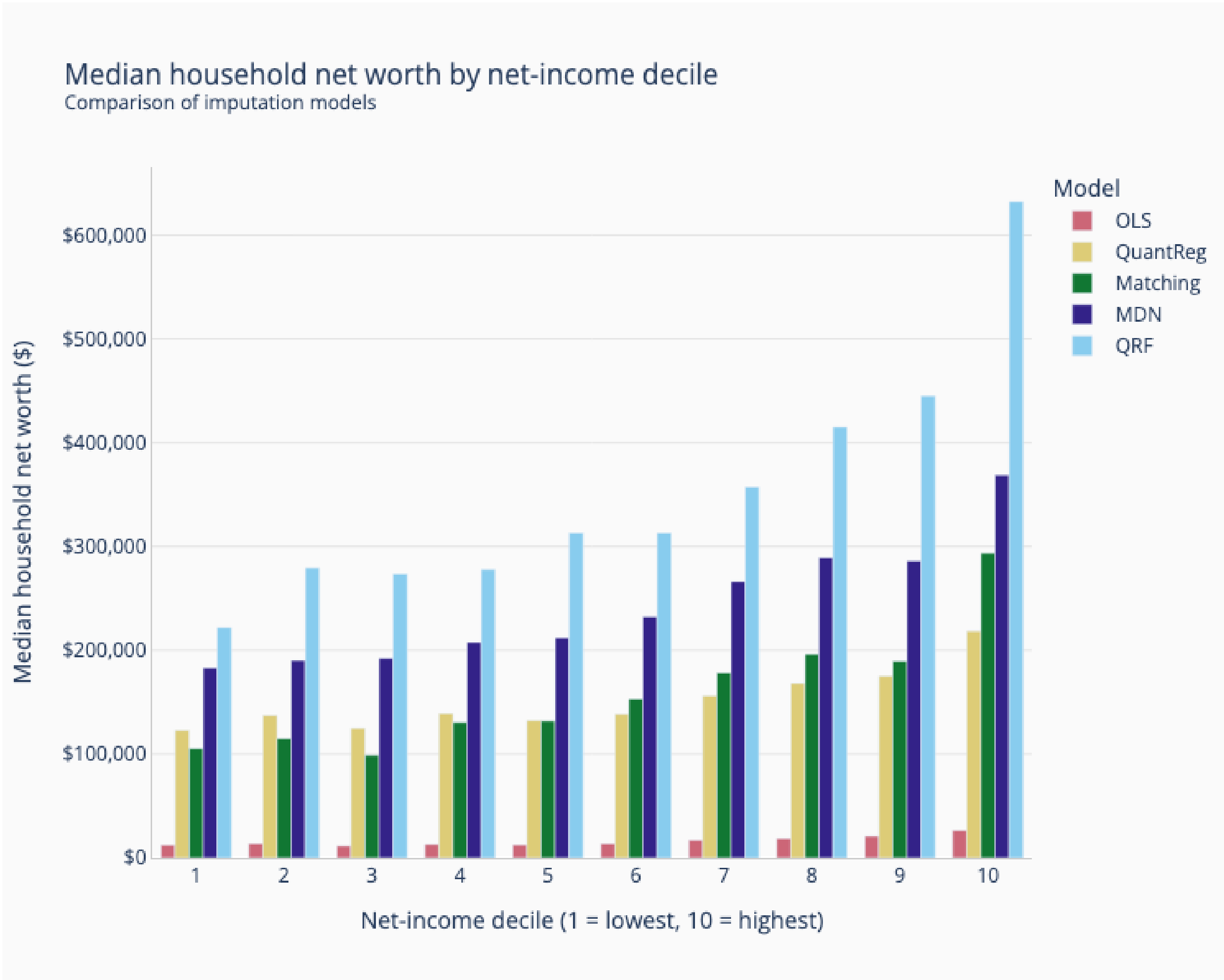

Figure 3: Average household net worth by disposable income decile model comparison. Households are divided into 10 equally sized groups based on their disposable income.

These results are consistent with the observations above. QRF produces a monotonically increasing average wealth across income deciles, as economic theory would predict. The plot also exposes weaknesses in the other models: OLS shows little variability in its predictions, and Matching consistently underpredicts across deciles.

5.2 Policy analysis: SSI simulation

As described in Section 4.4, we feed each model's imputed net worth into PolicyEngine as `ssi_countable_resources` and simulate SSI eligibility and benefits. Table 3 compares simulated recipient counts and total expenditure against SSA administrative totals for 2024.

Table 3: SSI simulation outcomes by imputation method compared to SSA administrative totals (2024). The baseline scenario uses PolicyEngine's default behavior, where `ssi_countable_resources` defaults to zero for all CPS households.

Model	Recipients (M)	Expenditure ($B)	Recipients % of Admin
Admin total	7.4	59.6	100.0%
Baseline (no wealth)	19.8	152.6	267.2%
QuantReg	8.3	74.6	111.7%
MDN	9.1	79.6	122.6%
Matching	11.1	90.2	150.2%
OLS	11.3	100.5	153.3%
QRF	12.1	102.8	163.3%

Without imputed wealth data, the baseline produces 19.8 million recipients and $152.6 billion in expenditure (267.2% of administrative totals). All five imputation methods substantially reduce this overestimation by screening out households whose imputed wealth exceeds the resource limits. Quantile Regression yields the closest match to administrative totals (111.7%), while QRF, despite achieving the best quantile loss, produces higher SSI estimates (163.3%). This divergence reflects the interaction between imputation accuracy and the policy function: QRF faithfully reproduces the conditional distribution of wealth, including the substantial mass of near-zero values in the SCF, which places more households just below the eligibility threshold. QuantReg's linear compression of the wealth distribution, while statistically less accurate, coincidentally pushes more values above the threshold. The remaining overestimation across all methods reflects that SCF net worth includes assets excluded from SSI countable resources (such as the primary residence and one vehicle).

5.2.1 Policy reform: SSI Savings Penalty Elimination Act

To show why wealth imputation matters for policy analysis, not just model calibration, we simulate the SSI Savings Penalty Elimination Act (119th United States Congress, 2025), which would raise resource limits from $2,000/$3,000 to $10,000/$20,000. Table 4 reports SSI outcomes under both current law and the reformed thresholds for each imputation method.

Table 4: SSI Savings Penalty Elimination Act: simulated reform impact by imputation method. "Current" columns reproduce the current-law results from Table 3; "Reform" columns show outcomes under $10,000/$20,000 resource limits; "Additional" columns show the difference. The baseline (no wealth) scenario produces zero reform impact because all resources default to zero under current law.

	Recipients (M)		Expenditure ($B)		Change	
Model	Current	Reform	Current	Reform	+Recip (M)	+Exp ($B)
QuantReg	8.3	8.4	74.6	75.6	0.1	1.0
MDN	9.1	9.8	79.6	84.5	0.7	4.8
Matching	11.1	13.0	90.2	100.6	1.9	10.4
OLS	11.3	15.6	100.5	124.6	4.2	24.1
QRF	12.1	12.7	102.8	106.2	0.6	3.4
Baseline	19.8	19.8	152.6	152.6	0.0	0.0

The reform raises the resource threshold from $2,000 to $10,000 (individual), so the relevant question is how dense each method's imputed wealth distribution is in the $2,000–$10,000 range. Households with imputed net worth in this interval fail the current-law resource test but would pass under the reform, gaining SSI eligibility. The estimated number of additional recipients ranges from 0.1 million (QuantReg) to 4.2 million (OLS), with corresponding additional expenditure from $1.0 billion to $24.1 billion. This range reflects how differently each imputation approach populates this critical region of the wealth distribution.

QRF estimates a modest reform impact (0.6 million additional recipients, $3.4 billion additional expenditure), consistent with its faithful reproduction of the SCF wealth distribution. Most households with near-zero wealth are already below the $2,000 threshold, so raising it to $10,000 captures relatively few additional households. OLS, by contrast, produces the largest reform impact (4.2 million additional recipients), because its compressed, normally distributed imputations place a substantial mass of households in the $2,000–$10,000 range. QuantReg's near-zero reform impact (0.1 million) reflects its tendency to push imputed values away from this intermediate range. These differences show that the choice of imputation method is not merely a technical detail but a modeling decision that directly shapes policy impact estimates.

The baseline (no wealth data) scenario shows plainly why imputation is necessary. Because all resources default to zero, every categorically eligible individual already passes the current-law resource test, so raising the thresholds has zero effect. Without wealth data, the microsimulation cannot distinguish between the status quo and a fivefold increase in resource limits.

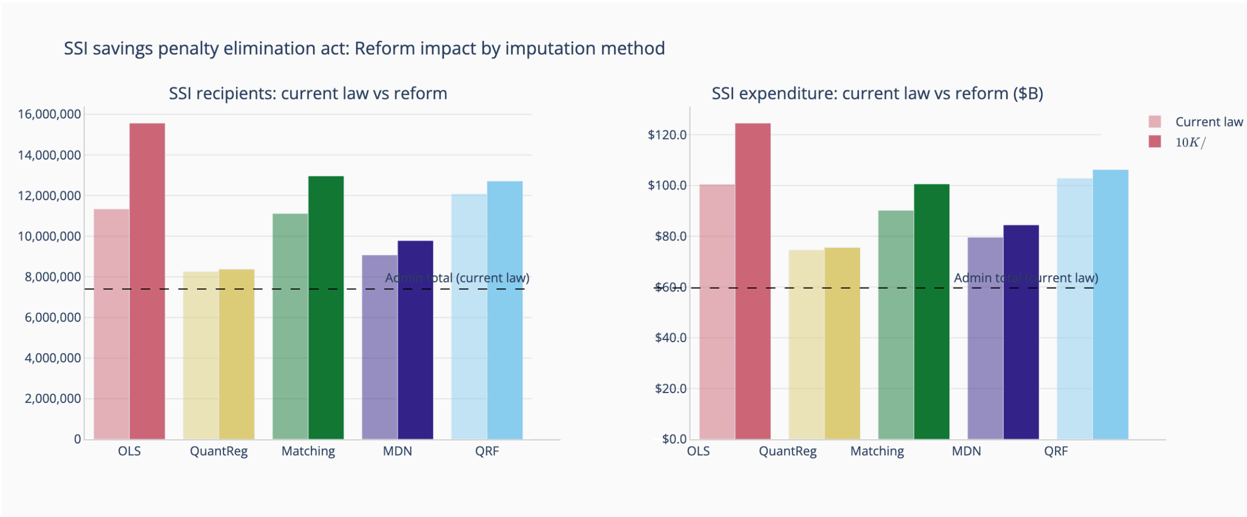

Figure 4: SSI recipients (left) and total expenditure (right) under current law (lighter bars) versus the SSI Savings Penalty Elimination Act reform (darker bars), by imputation method. The dashed line indicates the SSA administrative total under current law.

6 Discussion

This paper has evaluated five imputation methods within the `microimpute` framework, finding that Quantile Random Forests offer meaningful advantages for the SCF-to-CPS wealth imputation application. By preserving the full conditional distribution of wealth, QRF captures statistical properties of wealth data that traditional methods miss.

6.1 The role of `microimpute` in the analysis

Several design choices in `microimpute` shaped the analysis and its results. Wrapping all five methods behind a consistent API meant that performance differences in the benchmarking reflect the methods themselves, not differences in how they were implemented or tuned (PolicyEngine, 2025). The `autoimpute` function automated hyperparameter tuning and method selection based on quantile loss, removing a subjective step that could otherwise bias the comparison. Using quantile loss rather than RMSE as the evaluation metric was itself consequential: its asymmetric penalty structure prioritizes accuracy at distribution tails, where symmetric metrics are insensitive to the kinds of errors that matter most for skewed variables like wealth.

Two practical aspects also mattered. First, native survey weight support through stratified sampling allowed the models to train on data representative of the U.S. population despite the SCF's deliberate oversampling of wealthy households. Second, the full cross-validation pipeline, including hyperparameter tuning on the SCF for the QRF, MDN, and Matching models ran in under 45 minutes on standard hardware, making iterative experimentation practical. The package is open-source, so all algorithms and evaluation procedures can be inspected and modified (PolicyEngine, 2025).

6.2 Generalizability and predictor sensitivity

Two supplementary analyses help contextualize the main SCF-CPS results.

The cross-dataset benchmarking exercise (Appendix A) shows that method performance is context-dependent. QRF achieves the best Wasserstein distance on datasets with complex non-linear relationships, such as the Brazilian houses dataset and the SCF wealth imputation. Matching, by contrast, achieves the lowest Wasserstein distance on four of six benchmarking datasets and the best mean rank overall (1.67 vs. QRF's 1.83). This strong marginal performance follows from Matching's mechanism: drawing imputed values directly from the donor pool inherently preserves the shape and tail behavior of the original distribution, yielding low Wasserstein distances even when the conditioning on predictor values is suboptimal. But as the main SCF-CPS results show, good marginal distributions do not guarantee accurate conditional imputation. Households may receive plausible-looking wealth values that do not correctly reflect their individual characteristics. The trade-off between marginal and conditional accuracy, and the effect of differing survey designs on record matching, deserve careful attention in any application.

OLS performs competitively on datasets where the predictor-target relationship is approximately linear (e.g., abalone, space_ga), confirming that simpler parametric methods remain effective when their structural assumptions hold. MDN consistently ranks last across the benchmarking datasets, likely reflecting a combination of the relatively small sample sizes in these datasets and the hyperparameter sensitivity of neural network-based approaches, which require larger training sets to realize their flexibility advantage.

No single method dominates across all settings. The best choice depends on dataset size, the complexity of the predictor-target relationship, and whether the research question calls for conditional accuracy or marginal distributional fidelity. This is precisely why automated comparison matters: researchers should test methods on their own data rather than defaulting to any one approach.

The predictor importance analysis (Appendix B) reveals a clear hierarchy among the variables shared between the SCF and CPS. Financial predictors, particularly interest and dividend income, employment income, and age, account for the vast majority of imputation accuracy, with interest and dividend income alone more than doubling the quantile loss when removed. Demographic variables such as race and gender contribute relatively little to imputation quality. This concentration of predictive power in financial variables has two practical implications. First, the quality of wealth imputation rests primarily on the availability and accuracy of income-related variables in both surveys; researchers lacking financial predictors would see substantially degraded results. Second, diminishing returns from additional demographic predictors suggest that a parsimonious set of well-chosen financial variables can achieve most of the attainable accuracy, as confirmed by the progressive inclusion analysis showing that the first three predictors capture the majority of achievable performance. That said, demographic variables still provide incremental improvements and should not be excluded without reason.

The predictor importance and progressive inclusion tools in `microimpute` let researchers assess which variables drive imputation performance in their specific context and make informed decisions about variable selection when data are limited.

Together, these supplementary analyses put the main results in perspective. QRF's strong performance in the SCF-CPS setting reflects the particular characteristics of the wealth imputation problem, namely a large donor dataset, highly non-linear relationships, and informative financial predictors available in both surveys. Researchers applying `microimpute` to other settings should expect method rankings to vary with dataset characteristics.

6.3 Reform analysis and policy uncertainty

The SSI Savings Penalty Elimination Act simulation shows that imputation is not merely a data preprocessing step but a modeling choice with direct consequences for policy analysis. The variation in estimated reform impact across the five methods represents practical policy uncertainty: policymakers relying on different imputation approaches would reach different conclusions about the budgetary and coverage effects of raising SSI resource limits. This finding extends beyond SSI to any means-tested program where eligibility depends on wealth or asset data not collected in the primary survey.

By comparing current-law and reform outcomes within each method, the microsimulation approach (under ceteris paribus) partially mitigates the limitation that SCF net worth is only a proxy for SSI countable resources. Since the systematic bias affects both scenarios similarly, the reform impact estimates are less sensitive to this proxy than the level estimates. Nonetheless, future work could improve these estimates by imputing asset-specific components (e.g., excluding primary residence equity) rather than total net worth, and by extending the analysis to demographic subgroups (aged vs. disabled recipients) to better characterize the reform's distributional effects.

A practical path to resolving the variation in aggregate estimates across imputation methods is survey weight calibration. Rather than choosing a method based on proximity to administrative totals, which conflates imputation accuracy with policy function artifacts, researchers can select the method with the best conditional accuracy (here, QRF) for record-level imputation and then calibrate survey weights to match known administrative aggregates such as SSI recipient counts and total expenditure to ensure national-level representativeness (Woodruff and Ghenis, 2024). This two-stage approach combines accurate household-level wealth distributions that preserve the conditional relationships needed for reform simulation with aggregate estimates that align with administrative benchmarks.

6.4 Limitations and future improvements

6.4.1 Current package limitations

The current method library of five approaches could be expanded with gradient boosting machines, deeper neural architectures, or other modern machine learning methods, which may improve performance for complex multivariate relationships (Alaa et al., 2024). Ensemble methods that combine multiple imputation approaches across different parts of the distribution are another natural extension. Model selection and assessment could also benefit from evaluation metrics beyond quantile loss, giving a more complete picture of each model's behavior.

6.4.2 QRF-specific challenges

The terminal node sparsity issue identified in Section 2.4.5 remains a fundamental limitation of tree-based methods. When few training samples reach certain terminal nodes, multiple observations in the recipient dataset may receive identical imputed values, potentially underestimating variability in extreme wealth categories. Future work could explore adaptive tree construction methods that ensure minimum node occupancy or hybrid approaches that combine QRF with parametric methods at distribution extremes.

6.4.3 Missing data assumptions

All imputation methods in this study operate under the assumption that wealth is missing at random (MAR) conditional on the shared predictors X, as formalized in Section 2.2. This assumption may be violated in practice: wealthy individuals may be less likely to participate in surveys or may underreport assets, introducing missing-not-at-random (MNAR) patterns that the conditioning variables cannot fully account for (Andridge and Little, 2010). The predictor exclusion analysis in Appendix B provides a partial sensitivity check on the plausibility of the MAR assumption. The substantial degradation in imputation quality when financial predictors are removed, particularly the 116% increase in quantile loss when interest and dividend income is excluded, indicates that the conditioning set is doing meaningful work in explaining wealth variation, consistent with MAR plausibility. However, this analysis cannot rule out residual dependence on unobserved factors. More formal MNAR sensitivity analyses, such as pattern-mixture models or selection models applied to the imputation framework, are a natural next step. Researchers applying `microimpute` to settings where nonresponse or selection effects are suspected should consider supplementary diagnostics to assess the sensitivity of their results to departures from MAR.

These extensions would broaden the range of problems `microimpute` can address while preserving its current simplicity.

7 Conclusion

This paper has introduced `microimpute`, an open-source Python package for cross-survey imputation, and applied it to a substantive policy problem: imputing household wealth from the SCF onto the CPS. Comparing five imputation methods with an inverse hyperbolic sine transformation to accommodate negative and zero net worth values, we find that Quantile Random Forests achieve the strongest conditional accuracy for wealth imputation, owing to their ability to model the non-linear relationships between demographic predictors and wealth. A downstream SSI simulation confirms the practical importance of method choice. Without wealth data, the microsimulation overestimates SSI recipients by 167%, while imputation-based approaches substantially narrow this gap. Simulating the SSI Savings Penalty Elimination Act, which would raise resource limits from $2,000/$3,000 to $10,000/$20,000, further shows that method choice directly affects policy estimates. Reform impact estimates range from 0.1 to 4.2 million additional recipients and $1.0 to $24.1 billion in additional expenditure depending on the imputation method. Without imputed wealth, this reform cannot be simulated at all.

The findings also reveal nuances about method selection. Cross-dataset benchmarking shows that QRF's advantage reflects the specific characteristics of the wealth imputation problem rather than a universal superiority, and predictor importance analysis confirms that financial variables drive the majority of imputation accuracy.

Two broader implications follow. First, imputation method selection should be empirically driven rather than prescribed; the optimal approach depends on dataset characteristics, and standardized comparison tools are needed to support this. Second, the gap between statistical and policy-relevant evaluation metrics (QRF achieves the best quantile loss yet not the closest SSI estimates to administrative totals) points to the importance of validating imputation quality through downstream applications, not only through statistical benchmarks. Future work should expand `microimpute`'s method library to include gradient boosting and deep learning approaches (Alaa et al., 2024), explore ensemble strategies that combine methods across different parts of the distribution, and address the terminal node sparsity problem that limits tree-based methods at extreme quantiles.

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A Cross-Dataset Benchmarking

To validate `microimpute` across diverse domains beyond the SCF-CPS wealth imputation use case, we conduct a cross-dataset benchmarking exercise using six publicly available regression datasets from the OpenML AutoML Benchmark regression suite (Vanschoren et al., 2014), spanning different domains, sample sizes, and dimensionalities.

A.1 Datasets

Table 5 summarizes the datasets used in this analysis.

Table 5: Datasets used for cross-dataset benchmarking.

Dataset	Domain	Rows	Target Variable
space_ga	Voting/demographics	3,107	ln(VOTES/POP)
elevators	Aircraft control	16,599	Goal
brazilian_houses	Real estate	10,692	total_(BRL)
onlinenewspopularity	News/media	39,644	shares
abalone	Marine biology	4,177	Class_number_of_rings
house_sales	Real estate	21,613	price

A.2 Methodology

For each dataset, we split the data 60/40 into donor and receiver subsets. The `autoimpute` pipeline runs internally on the 60% donor portion (cross-validation for method selection), and the selected method then imputes onto the 40% receiver subset. We evaluate imputation quality using the Wasserstein distance between the imputed and true distributions on the receiver subset. We use Wasserstein distance rather than quantile loss for this benchmarking because it provides a single summary measure of marginal distributional accuracy, which is more directly comparable across datasets with different scales and distributional shapes.

A.3 Results

Table 6 reports the Wasserstein distance for each method on each dataset, with lower values indicating better distributional recovery.

Table 7 summarizes the mean rank of each method across all six datasets.

Matching achieves the best Wasserstein distance on four of six datasets and the lowest mean rank overall (1.67), followed closely by QRF (1.83). QRF performs best on space_ga and brazilian_houses, the latter of which exhibits complex nonlinear relationships between predictors and house prices. OLS and Quantile Regression occupy middle ranks, while MDN consistently ranks last, likely reflecting a combination of the relatively small sample sizes in these datasets and the hyperparameter sensitivity of neural network-based approaches.

With only six datasets, the rank differences between methods, particularly the narrow gap between Matching (1.67) and QRF (1.83), should be interpreted cautiously; they are

Table 6: Wasserstein distance by method and dataset. Bold values indicate the best-performing method for each dataset.

Dataset	Matching	QRF	OLS	QuantReg	MDN
space_ga	0.011	**0.010**	0.027	0.098	0.050
elevators	**0.0002**	0.0003	0.001	0.002	0.020
brazilian_houses	120.5	**27.2**	99.4	112.4	3,378.6
onlinenewspopularity	**239.9**	783.7	2,777.1	2,376.8	2,038.0
abalone	**0.003**	0.003	0.004	0.003	0.040
house_sales	**8,797.8**	12,576.6	41,862.3	64,725.3	544,730.4

Table 7: Mean rank across all six benchmarking datasets (lower is better).

Method	Mean Rank
Matching	1.67
QRF	1.83
OLS	3.33
QuantReg	3.67
MDN	4.50

not statistically robust to the inclusion or exclusion of individual datasets. The relative rankings here also differ from the main SCF-CPS analysis, where QRF achieves the lowest quantile loss. This is consistent with the expectation that method performance depends on dataset characteristics, and it supports the case for empirically comparing methods on each new dataset rather than relying on a single approach.

B Predictor Importance Analysis for SCF-CPS Imputation

To assess the relative contribution of each predictor variable to the quality of wealth imputation, we conduct two complementary analyses using the QRF model on the SCF data: a leave-one-out analysis and a progressive inclusion analysis. These analyses use six of the seven predictor variables from the main experimental setup; `own_children_in_household` is excluded because it was not available in the SCF preprocessing for this robustness analysis. The results help evaluate which shared variables between the SCF and CPS are most informative for imputing net worth.

B.1 Leave-one-out analysis

We measure the impact of each predictor by excluding it from the QRF model and observing the resulting increase in average quantile loss relative to the full model (baseline loss:

1,661,723, computed on the original dollar scale rather than the asinh-transformed scale used in Section 5). Table 8 reports the results.

Table 8: Leave-one-out predictor importance for QRF wealth imputation. Relative impact measures the percentage increase in average quantile loss when the predictor is excluded.

Predictor Removed	Avg Quantile Loss	Loss Increase	Relative Impact
interest_dividend_income	3,596,108	1,934,385	116.4%
employment_income	2,219,377	557,654	33.6%
age	2,189,845	528,122	31.8%
pension_income	1,875,120	213,397	12.8%
race	1,728,080	66,358	4.0%
is_female	1,698,452	36,729	2.2%
Baseline (all predictors)	1,661,723	—	—

Interest and dividend income is by far the most important predictor, with its removal more than doubling the quantile loss (116.4% increase). Employment income and age contribute substantially as well (33.6% and 31.8% increases, respectively). Pension income has a moderate effect (12.8%), while race and gender contribute relatively little to imputation accuracy (4.0% and 2.2%).

B.2 Progressive inclusion analysis

We complement the leave-one-out results by adding predictors one at a time, following the importance ordering determined by the leave-one-out analysis (Table 8), and measuring the marginal improvement at each step. Table 9 reports the results.

Table 9: Progressive predictor inclusion for QRF wealth imputation. Predictors are added in order of importance, and marginal improvement measures the reduction in average quantile loss from adding each predictor.

Step	Predictor Added	Avg Quantile Loss	Marginal Improvement
1	interest_dividend_income	5,037,628	—
2	age	2,752,098	2,285,529
3	employment_income	1,932,222	819,876
4	pension_income	1,737,197	195,025
5	race	1,704,094	33,103
6	is_female	1,685,442	18,652

The progressive analysis confirms the leave-one-out findings. The first three predictors (interest/dividend income, age, and employment income) account for the majority of imputation accuracy, with diminishing returns from additional demographic variables. All six predictors together yield the lowest quantile loss (1,685,442), indicating that each variable contributes positively, though the marginal gains from race and gender are small. The small difference between this value and the leave-one-out baseline (1,661,723) reflects variation across cross-validation fold assignments and random seeds between the two analyses.

B.3 Implications for the Conditional Independence Assumption

The strong predictive power of the shared financial variables, particularly interest and dividend income, employment income, and pension income, is consistent with, though not sufficient evidence for, the plausibility of the CIA in the SCF-CPS matching context. These variables capture the main dimensions of household financial position available in both surveys. The fact that they collectively explain a large share of the variation in net worth suggests that the common variables X may be sufficient to render the target variable Y (wealth) approximately conditionally independent of variables Z unique to the CPS, though this assumption remains fundamentally untestable as discussed in Section 2.2.